

RESEARCH ARTICLE

Distributed Anomaly Detection Using Federated Learning

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ABSTRACT One of the major challenges video surveillance systems usually face is high communication costs and computational bottlenecks, leading to latency in the transfer of high-definition video data. Such issues generally arise when large-scale projects are implemented in real-time capabilities. The paper explores Federated Learning (FL) algorithms for an efficient and scalable suspicious activity detection system to address these limitations. Federated Learning allows a decentralized approach that allows model training on participating devices (clients), and only the model updates are shared with the central server instead of sharing a huge amount of data. The approach reduces bandwidth consumption and minimizes latency without compromising the performance of the model. This study is among the first to explore federated learning (FL) for video-based anomaly detection using the UCF-Crime dataset. We propose and comparatively evaluate three FL aggregation strategies—FedAvg, Unweighted FedAvg, and FedSGD—under realistic, non-IID client data distributions. Unlike existing centralized methods, our approach offers enhanced privacy and reduced communication overhead while maintaining competitive performance. The results obtained in the experiments conducted on a varying number of clients for each approach suggest Unweighted FedAvg as the most reliable approach, which consistently achieves the highest accuracy and lowest loss for any given number of clients. The Accuracy for FedAvg ranges from 0.73 for 5 clients to 0.82 for 30 clients, and the loss decreases from 0.81 for 5 clients to 0.64 for 30 clients. Similarly, Unweighted FedAvg shows a consistent increase in accuracy from 0.74 for 5 clients to 0.85 for 25 clients and then drops slightly to 0.84 for 30 clients, and similarly for loss also it decreases from 0.78 for 5 clients to 0.53 for 25 clients and increases slightly to 0.56 for 30 clients. Despite the slight decrease in accuracy and increase in loss, it always performs better than FedAvg. Whereas FedSGD has shown a very erratic pattern in change of accuracy and loss as the number of clients increases, with the lowest accuracy being 0.29 and the highest being 0.41, and for loss highest was 1.91 and the lowest was 1.59. The study successfully uncovers the potential of federated learning to address the limitations, such as computational bottlenecks, latency in data transfer, and model performance over non-iid data in real-time implementations of the public security systems.

INDEX TERMS Federated learning, video surveillance, anomaly, distributed learning, detection.

I. INTRODUCTION

With the shift in monitoring paradigms, technologies such as continuous log generation, video surveillance [1], [2], and sensor-based monitoring have proven to be highly effective for monitoring various environments. The primary objective

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of these systems is to monitor and interpret the input data. In the case of video surveillance, this task is often performed manually [3]. However, replicating human intelligence in interpreting complex scenarios remains challenging, making the automation of this process a demanding task [4]. Various machine learning algorithms can be employed to analyze video streams, including Support Vector Machines (SVM) [5], Convolutional Neural Networks (CNN) [6],

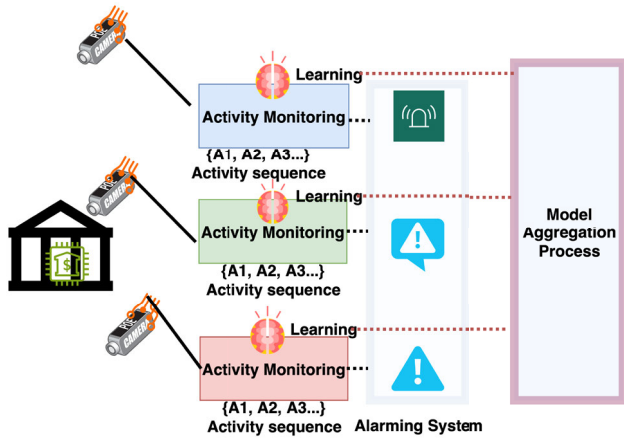


FIGURE 1. Federated learning-based activity monitoring system with decentralized training and aggregation.

and Long Short-Term Memory (LSTM) networks [7]. Deep learning models have effectively detected suspicious activities in real-time environments [8]. Nevertheless, challenges such as network latency and outages, hardware failures, and the need to ensure continuous data availability persist. Continuous data availability typically refers to the uninterrupted and timely access to relevant data sources needed for monitoring and analysis.

One of the efficient approaches is to stream all data to the cloud for processing [3] and detection of suspicious activities. However, this method incurs high communication costs and introduces additional security concerns.

To address the challenges of transferring data to centralized systems for processing, federated learning emerges as a promising alternative. In federated learning, data is processed locally, and only the relevant information is shared with a central system [9]. This information is aggregated to generate a global model. Federated learning allows the training of a shared global model collaboratively by multiple clients or nodes, utilizing the data stored locally on the nodes [10].

The implementation of federated learning for detecting suspicious activities holds significant potential. Figure 1 illustrates a basic conceptual framework for applying the federated learning approach to the detection of suspicious activities. The traditional machine learning approaches for surveillance typically rely on centralized systems that collect and process large volumes of video data on central servers. The centralized models lead to significant issues in the transfer of high-definition videos, usually resulting in latency, leading to inefficiencies and delays. Apart from this bottleneck, the extensive computational load required for training models on a single server can significantly affect the efficiency of the centralized infrastructure, especially in large-scale surveillance environments. Federated Learning (FL) addresses this challenge by distributing the computational load across multiple clients, allowing local processing of data while only sharing model updates

TABLE 1. Notations and their description used in the proposed approach.

Nation	Description
w^0	Initial Model
w^r	Global Model after r^{th} round
w_i^r	Updated model from client i after r^{th} round
N	Number of clients in the network
n	Number of clients participating in a round
D_i	Number of data points at client i
d_i	Local data at client at client i
x_k^i	Feature vector representing video frame
y_k^i	Corresponding label for the feature vector
BW	Bandwidth available for data transfer
R	Resolution of the video
F	Frame rate in frames per second
B	Bits per pixel
T	Duration of video in seconds of a client
$ w^r $	Size of model updates
α_i	Client specific weighting factor
g_i	Gradient descent of the loss function at client i
K_i	Local dataset of client i
∇	Gradient Operator
g_r	Aggregated SGD after r^{th} round.

(weights/gradients) with a central server. This reduces the need for high-bandwidth transmission, minimizes latency, and ensures that computational resources are optimized, making the real-time suspicious activity detection process more efficient and scalable.

The novelty of this work lies in applying federated learning to anomaly detection in video surveillance, an area where most existing methods remain centralized. To the best of our knowledge, this is one of the first studies to evaluate FL strategies on the UCF-Crime dataset, using a realistic, non-IID data distribution across simulated clients. We compare FedAvg, Unweighted FedAvg, and FedSGD to assess performance and communication efficiency in privacy-sensitive, distributed environments. Our work also includes a communication cost analysis specific to video surveillance—a dimension not previously addressed in similar FL research.

II. LITERATURE REVIEW

The increasing ubiquity of surveillance systems in public spaces has prompted significant advancements in computer vision and machine learning for detecting anomalous or suspicious activities. Traditional surveillance relied heavily on human operators, which not only limited scalability but also introduced inconsistencies due to human fatigue and subjective interpretation. Over the past two decades, research has transitioned toward automated systems capable of real-time threat analysis and behavior detection, facilitated by developments in deep learning, edge computing, and, more recently, federated learning. Early works focused on constructing foundational systems for video surveillance. Elharrouss et al. [1] provided a comprehensive overview of video surveillance architectures, exploring the evolution from analog to intelligent surveillance systems capable of real-time analysis. These systems laid the groundwork by establishing the importance of both spatial and temporal features in

interpreting human activities. The emergence of human fall detection techniques, like the hybrid Haar cascade classifier introduced by Kaur et al. [2], showcased initial attempts to apply classical computer vision methods in real-time safety-critical applications. Though effective in constrained environments, such rule-based approaches often lacked robustness when applied to unstructured, crowded, or dynamic real-world settings. To address this limitation, several researchers pivoted towards data-driven methods. Loganathan et al. [3] and Bhugul-Rajurkar and Gulhane [4] proposed suspicious activity detection pipelines that employed basic statistical analysis or hand-crafted feature extraction combined with classifiers. However, these approaches were typically limited by their dependence on manually engineered features and their inability to generalize across diverse surveillance scenes. As computational capabilities increased, the field witnessed a paradigm shift toward machine learning models, particularly support vector machines (SVMs) [5] and later, convolutional neural networks (CNNs) [6], which were better suited for capturing intricate visual patterns from raw data. With the maturation of deep learning, more sophisticated anomaly detection frameworks began to emerge. Nassif et al. [7] provided a systematic review of machine learning techniques for anomaly detection, emphasizing the versatility of unsupervised and semi-supervised learning in handling the scarcity of labeled data—a prevalent challenge in surveillance applications. Building on this, Qasim and Verdu [8] integrated CNNs with recurrent neural networks (RNNs) to harness both spatial and temporal dynamics of video frames, highlighting the need for sequence modeling in real-world anomaly detection systems. Despite these advancements, centralized learning approaches presented several critical challenges, particularly concerning data privacy, bandwidth constraints, and the risk of centralized failure. More recently, continual learning (CL) and its federated counterpart, federated continual learning (FCL), have emerged as key strategies for adapting models to evolving data streams without retraining from scratch. CL aims to mitigate catastrophic forgetting when new information arrives over time, while FCL integrates this capability within distributed environments. Notable examples include Ma et al. [11], who proposed a knowledge-distillation-based FCL method, and Faber et al. [12], who developed a collaborative replay system for intrusion detection in distributed systems. These directions highlight the growing need for adaptable and privacy-preserving learning in dynamic settings, particularly relevant to anomaly detection in long-term surveillance systems. Federated learning (FL) emerged as a compelling alternative by enabling model training directly on edge devices while maintaining data locality. The work of Nilsson et al. [9] was pivotal in benchmarking various FL algorithms, evaluating their performance and convergence behavior. Xia et al. [10] further contextualized FL in the edge computing landscape, identifying several open research questions, including device heterogeneity, communication efficiency, and security vulnerabilities. In the

realm of surveillance, researchers have explored how deep learning could be utilized to detect crime-like or suspicious behaviors directly from video footage. Kolaib and Waleed [13] employed a deep learning-based architecture to model crime patterns, utilizing spatiotemporal information, though their approach was still rooted in a centralized framework. The real-world implications of anomaly detection were more thoroughly explored by Sultani et al. [14], whose work proposed a robust dataset and anomaly detection methodology for untrimmed surveillance videos. Their findings underscored the challenges of ambiguous activity boundaries and the context-dependent nature of anomalies, further complicating training processes. Complementary to this, Gulati et al. [15] proposed using CNNs combined with a rolling average method to smooth prediction volatility and increase stability in behavior classification, especially in low-resolution footage. Patwal et al. [16] and Yuan et al. [17] emphasized the semantic understanding of human activities in videos, integrating vision-language models to elevate interpretability—a crucial factor in security-critical applications. The need to infer intent or suspiciousness, rather than merely recognizing motion, represents a crucial frontier in surveillance AI. Recent studies have begun to leverage deeper architectures for improved context modeling. Kansal et al. [18] introduced SUSPACT, a real-time system using deep learning for predicting suspicious behavior. Their work attempted to bridge the gap between theoretical models and practical implementation, particularly in environments like airports or metro stations. Similarly, Kumar and Biswas [5] implemented a hybrid of CNN and recurrent models, enhanced with fuzzy logic, to better interpret uncertain or ambiguous human behaviors. These models underscore a recurring trend: combining multiple modalities (vision, semantics, and temporal data) often leads to better generalization and interoperability. Parallel efforts such as those by Dwivedi et al. [19] and Kumar et al. [20] focused on enhancing temporal modeling using attention mechanisms and bidirectional LSTMs. Such architectures allow the system to factor in both past and future contextual cues, which is especially valuable for early anomaly detection. These studies further emphasized the growing reliance on sequence-based modeling in understanding human behavior, particularly in surveillance footage with complex temporal dependencies. On a different front, person re-identification has become a tool not just for tracking but also for inferring anomalous patterns through trajectory analysis. Rehman et al. [21] combined person re-identification with behavior profiling to flag potential threats, leveraging multi-camera networks and spatiotemporal signatures. This development signals a shift toward contextual intelligence in surveillance, where the suspiciousness of an activity is partly derived from the identity and historical behavior of the individual. Moreover, systems such as the drone-based CNN surveillance proposed by Vuyyuru et al. [22] highlight the intersection of mobility and intelligent surveillance. By incorporating enhanced feature selection techniques,

the model demonstrated higher precision in dynamic outdoor environments—areas previously underrepresented in static camera-based datasets. The concept of federated learning has been further refined to fit anomaly detection tasks. For instance, Collins et al. [23] discussed the fine-tuning of FedAvg for improved representation learning, addressing a critical issue in FL: model drift due to non-IID data. In parallel, Balik [24] compared Federated SGD and FedAvg in a healthcare context, providing insights that are also applicable to surveillance, especially when device-level data distributions vary significantly. These studies emphasize that while FL holds promise, its naive implementation may not yield optimal results in all contexts—especially those with heterogeneous surveillance data across edge nodes. Anomaly detection has also benefited from advanced graph-based models. Khan and Ebrahim [25] introduced ANOGAT-Sparse-TL, which combines graph attention with sparsification to detect anomalies in attributed networks. While not directly applied to surveillance videos, the framework's strength in representing relational data could inspire adaptations for modeling interactions between individuals in a monitored scene. Similarly, sparse canonical correlation analysis and residual-enhanced graph convolutions [26], [27] demonstrate how structural information (e.g., social graphs or inter-object relationships) can be integrated to uncover hidden anomalies, particularly in complex environments where group dynamics influence behavior. Finally, the work by Khan et al. [28] on anomalous node detection using dual variational autoencoders and GANs signals an important methodological development: the fusion of generative models with robust latent space encoding to improve anomaly detection. This approach holds promise for federated surveillance systems by providing flexible representations that can generalize across diverse edge devices while maintaining privacy through decentralized training. Despite these advancements, several challenges remain. First, the scarcity of labeled datasets reflecting real-world suspicious activities limits supervised learning efficacy. Second, the non-IID nature of surveillance data across multiple public spaces—due to differences in lighting, crowd density, camera angles, and local context—poses a major challenge for federated learning systems. Third, current systems often lack interpretability, a crucial feature in legal or law enforcement settings where system decisions must be auditable. In light of these gaps, integrating federated learning with advanced anomaly detection techniques appears as a promising frontier. Such integration not only ensures data privacy and scalability but also enables continuous model improvement across heterogeneous edge devices. Moreover, the potential for combining graph-based reasoning with temporal and spatial modeling—augmented by attention mechanisms and generative learning—can create a new class of intelligent, context-aware surveillance systems. As we progress, the emphasis should be on creating adaptable models that account for contextual cues, person-specific behavior histories, and evolving definitions of

‘suspicious activity.’ Such systems must be capable of real-time operation, resistant to adversarial attacks, and respectful of individual privacy. These goals necessitate further research in privacy-preserving AI, unsupervised learning, and distributed architectures—directions where federated learning and deep generative models are poised to play transformative roles.

III. PROBLEM FORMULATION

The real-world surveillance systems usually depend on traditional machine learning approaches for the detection of suspicious activities and respond in real time.

For a surveillance system of N devices where each device $i \in \{1, 2, \dots, N\}$ collects video data locally $d_i = (x_k^i, y_k^i)_{k=1}^{D_i}$ which remains on the local devices.

In traditional machine learning systems, the high definition video footage are sent from the N devices participating in the network to a central system for training.

For the local dataset d_i of client i with size:

$$|d_i| = D_i \cdot (\text{size}(x) + \text{size}(y)) \quad (1)$$

The communication cost of transmitting the complete video data from all N participating devices is:

$$C_{comm} = \sum_{i=1}^N |d_i| = \sum_{i=1}^N D_i \cdot (\text{size}(x) + \text{size}(y)) \quad (2)$$

In real-time scenarios for high-dimensional data, the size of x_k^i is large and can increase the communication overhead manifold. Traditional approaches can also result in scalability issues, as the increase in the number of clients can exponentially increase the communication cost of transferring the video data.

The traditional machine learning approaches introduce latency due to high bandwidth usage by high-definition video data being transferred from devices to the central server.

In centralized ML approaches, raw video data is transferred from each device to the central server, where:

Size of video data per device ($Data_{video}$):

$$Data_{video} = R.F.B.T \quad (3)$$

Hence, the total data transferred by N devices:

$$Data_{video_total} = N \cdot Data_{video} \quad (4)$$

and the transmission time taken to transfer the video data is:

$$T_{centralized} = \frac{Data_{video_total}}{BW} \quad (5)$$

To address the issue of high communication costs/overhead, one of the possible solutions would be to use federated learning, which encourages the sharing of model updates from the server to devices and vice-versa instead of actual video data.

Hence, the data transferred per round:

$$Data_{FL} = 2.N \cdot |w'| \quad (6)$$

2 represents data transfer in a complete round, i.e., client to server and server to client communication.

And, total data transferred over R rounds:

$$Data_{FL_total} = R.Data_{FL} = R.2.N.|w^r| \quad (7)$$

The transmission time taken to transfer the model updates over all the rounds of training is:

$$T_{FL_total} = \frac{Data_{FL_total}}{BW} \quad (8)$$

IV. PROPOSED APPROACH

The proposed approach is divided into four phases that include local data preprocessing and conversion on clients, initialization of a base/initial model that is shared with a subset of the clients participating in the round. Further, the initial model is trained by the individual clients, updating the model parameters. After local training, each client of the training round sends only the model updates to the central server for aggregation. The server aggregates these updates to create a global model, often using weighted averaging based on factors such as data quality or volume. This global model is shared again with a different client subset in the network for the next round of training, and the cycle continues.

Phase 1: Local Data Preprocessing and Conversion on Clients:

In the first phase, the raw data on the local clients' needs to be preprocessed. Each client begins by converting its local data into a standardized format. This is done by transforming each image into an ordered dictionary format, where the dictionary serves as a structured way to store individual training samples.

The label stores the class to which the individual frame belongs and helps the model learn to classify the activity, while the pixels provide the features of the image that the model processes. Each client holds its data D_k in an ordered dictionary format. This ordered dictionary format is essential for consistent data handling across all clients. After this, the data is saved as TFRecord files.

Phase 2: Model Initialization and Local Training:

The second phase involves initialization of the initial global model w^0 that will be distributed to the clients for the first round of training. The model initialized is a fully connected feedforward neural network with random weights assigned.

This initialized model is then sent to each of the participating clients. To improve the efficiency of the transmission, instead of sharing the complete model, the server shares only the model parameters (weights and biases) with the clients. These parameters represent the state of the global model. Usually, a predefined model architecture is used so that the server does not need to repeatedly share the architecture unless it changes. The type signature of the global model's weights and other parameters being shared with the clients is:

The central server initializes a global initial model w^0 . The initialization of the global model is usually done randomly, or can be pre-trained, or based on some previous knowledge.

The global model w_0 is then distributed to the participating clients of the round.

For N clients in the network and n clients participating in the round, each client $i \in \{1, 2, \dots, N\}$ trains the model parameters on its local data. The main aim for each client is to minimize the loss function defined on its data. The clients then share the locally updated model parameters w_i^r with the server after the r^{th} round.

Phase 3: Model Aggregation on the Central Server:

Once the clients have trained the global model on their local data and the server has received the model parameter updates from the clients, the server aggregates the model updates from all the participating clients. The central server aggregates the received updates using weighted averaging, which uses certain factors to decide the contribution of each client in the final updated model.

The updated model after the r^{th} round can be written as:

$$w^r = \sum_{i=1}^n \alpha_i \cdot w_i^r \quad (9)$$

where:

w_i^r : Updated weights from client i after local training round r .

n : Number of clients participating in the round.

α_i : A client specific weighting factor.

Once the aggregation is complete, the central server updates the global model with the averaged weights and sends the updated model back to the clients. This global model, now updated with knowledge from all local clients, serves as the starting point for the next round of local training.

$$w^{r-1} \leftarrow w^r \quad (10)$$

A. FEDAVG (FEDERATED AVERAGING)

References [23] and [24] Federated Averaging (FedAvg) is a popular algorithm used in federated learning to train a shared global model across multiple clients (devices), each with its local dataset. The algorithm updates the global model by averaging the local model parameter updates from all clients, weighted by the number of data points they have. The weighting factor in FedAvg, i.e. α_i , is the ratio of the number of data-points on client i (D_i) and total data-points on all the clients participating in the round ($\sum_{i=1}^n D_i$).

$$\alpha_i = \frac{D_i}{\sum_{i=1}^n D_i} \quad (11)$$

After training on the local data, clients share the updated model w_i^r to be aggregated onto the global model. The mathematical view of the aggregation looks like:

$$w^r = \sum_{i=1}^n \frac{D_i}{\sum_{i=1}^n D_i} w_i^r \quad (12)$$

Weighted Averaging: The contribution of each client's model to the global update is proportional to the number of local data points. This ensures that clients with more data have a stronger influence.

Communication Efficiency: Clients only send model updates, not raw data, preserving privacy and reducing communication costs.

Heterogeneity Handling: FedAvg can handle non-IID (non-independent and identically distributed) data across clients.

B. UNWEIGHTED FEDAVG

Whereas in FedAvg, the server aggregates the weights based on the size of the local dataset at the individual client. This ensures that the weights from the clients with a higher amount of local data have a large influence on the global model.

In the experiments, we also have implemented unweighted FedAvg to see if the size of the dataset at clients affects the performance of the model in terms of sparse categorical accuracy and overall loss. As the name suggests, in Unweighted FedAvg, all clients are treated equally, regardless of the size of their local datasets. The weights w_i^r from all the participating clients are aggregated at the server using an unweighted average.

$$w^r = \frac{1}{n} \sum_{i=1}^n w_i^r \quad (13)$$

C. FEDSGD FEDERATED STOCHASTIC GRADIENT DESCENT

Reference [24] FedSGD is a federated learning approach particularly used to address problems faced in Resource-Constrained Devices, Homogeneous Data Distribution, and Frequent Model updates. FedSGD helps to extend the traditional Stochastic Gradient Descent to a federated learning approach.

The server shares the initial global model w^0 with the participating clients. Each client i computes the gradient g_i of the loss function $l(w^r, K_i)$ concerning its local data, where K_i is the local dataset of client i .

For client i , the gradient is:

$$g_i = \nabla l(w^r, K_i) \quad (14)$$

The participating clients of the round share the computed gradients with the server. The server aggregates the gradients received from the clients. The gradients are the only information shared by the clients with the server. The server uses these aggregated gradients to update its initial model parameters. The aggregation is typically done using weighted averaging, and the weights depend on the size of the client's dataset.

$$g_r = \frac{\sum_{i=1}^n D_i \cdot g_i}{\sum_{i=1}^n D_i} \quad (15)$$

The server utilizes the aggregated gradient g_r to update the global model w^{r-1} after the r^{th} round with a learning rate of η to get the updated server model w^r .

$$w^r = w^{r-1} - \eta \cdot g_r \quad (16)$$

The process repeats itself for a pre-decided number of rounds on different subsets of clients or till the model converges.

TABLE 2. Distribution of images (frames extracted) across different classes.

S.No.	Class	Numer of Images
1	Abuse	19,373
2	Arrest	29,762
3	Arson	30,007
4	Assault	13,017
5	Burglary	47,161
6	Explosion	25,263
7	Fighting	25,915
8	Normal Videos	1,012,720
9	Road Accident	26,149
10	Robbery	42,328
11	Shooting	14,770
12	Shoplifting	32,458
13	Stealing	46,786
14	Vandalism	14,737

V. RESULTS AND ANALYSIS

To validate the proposed approach, a detailed analysis was conducted on the UCF crime dataset [14]. The experiments were conducted under consistent training conditions to enable a fair comparison between aggregation strategies. While this setup helps highlight performance differences across methods, we acknowledge that further evaluation across varied runs could strengthen the overall robustness assessment. The UCF Crime dataset is a wide-ranging dataset that contains 128 hours of 1900 untrimmed real-world surveillance videos that are categorized into 14 different classes. 13 being the anomalous activities and 1 being the normal events.

Table 2 gives an accurate view of the shape of the dataset. The table gives the exact number of frames extracted from the surveillance videos from each class. A sample of the data set is shown in Figure 2, i.e. example of frames extracted for each class. The experiments are conducted to address the following questions:

- R1: How does the proposed model adapt to the incremental addition of the data in terms of accuracy during the training and testing phases?
- R2: How does the proposed approach adjust to the loss during the training phase?
- R3: How does the proposed approach react to the varied number of clients?
- R4: A comparative analysis of the different aggregation methods.

A. EXPERIMENTAL SETUP

In the UCF Crime dataset, the surveillance videos are classified into 14 classes. Where one single class contains multiple videos. For the experiment, we have simulated 30 clients that receive a unique distribution of the data that represents different environments (real-world scenarios). This results in a highly heterogeneous dataset, i.e., a non-iid dataset where each client receives different classes and different frequencies of videos for each class.

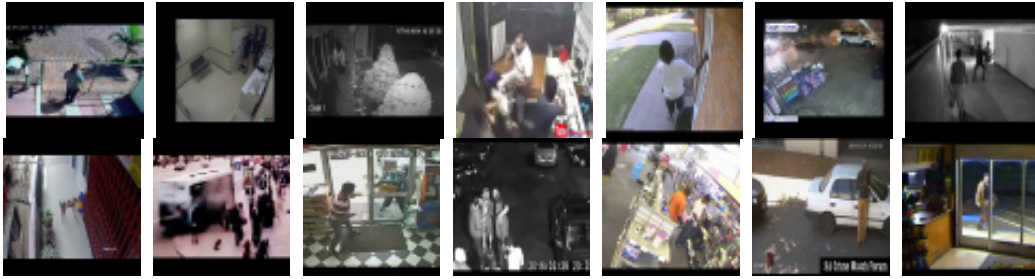


FIGURE 2. Sample surveillance images representing 14 distinct activity classes.

B. EXPERIMENTS DONE

Number of clients: This parameter specifies the number of client models. For example, if it is 30, that means there are 30 clients, each has its own model. However, the model shape for each client is the same.

Rounds: It means that if this factor is 50, then 50 times the client has sent the updated parameters to the server, and 50 times the aggregation of the model is done and communicated back to the clients. To record the results, i.e., Accuracy and loss, experiments have been conducted over a varying number of clients (5-30) each for 50 rounds. The recorded results have been utilized to understand the pattern in change of the metrics with change in the number of clients and how the proposed approaches perform concerning the traditional CNN. Table 3 provides a detailed performance comparison of the different aggregation approaches implemented over different numbers of clients for 50 rounds.

C. ACCURACY ANALYSIS

Table 3 indicates that accuracy and loss improve consistently for both FedAvg and unweighted FedAvg with increasing number of clients, while FedSGD performed poorly for both metrics and showed an inconsistent change in both metrics.

To provide a stronger context for evaluating the proposed federated learning approaches, we also implemented a centralized Convolutional Neural Network (CNN) model as a benchmark. This model was trained under similar experimental conditions using the same dataset (UCF-Crime) and serves as a non-federated baseline for performance comparison. The CNN was trained on the entire dataset in a centralized manner for 10 epochs, and its performance in terms of training accuracy, test accuracy, and loss is reported in Table 3 and Table 4.

The inclusion of the CNN model helps illustrate the trade-offs between centralized and federated approaches. While the centralized CNN achieves a relatively stable performance (test accuracy: 0.69, loss: 0.94), our federated models, particularly Unweighted FedAvg, show comparable or even improved accuracy in distributed environments, while significantly reducing communication costs and preserving data privacy. This reinforces the practical value of federated learning in real-world surveillance scenarios where centralized data aggregation is not feasible.

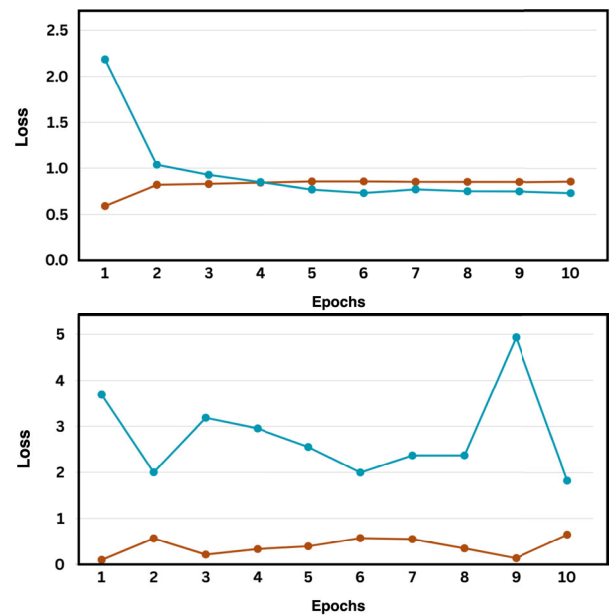


FIGURE 3. CNN model training: accuracy and loss trends over epochs for training and validation sets.

Though we can see that the centralized training with CNN table but the federated learning approaches can achieve better results when working with multiple devices in the network.

FedAvg, as shown in Figure 4 showed a consistent increase in accuracy that started at 0.73 for 5 clients to 0.82 for 30 clients, while reporting a test accuracy in Table 4 of 0.68 for 5 clients and a consistent increase up to 0.77 for 30 clients. In Figure 5, it also shows a consistent decrease in loss with 0.81 for 5 clients and dropping to 0.64 for 30 clients. Whereas Unweighted FedAvg shows a consistent increase in accuracy in Figure 8, from 0.74 to 0.85 and from 0.69 to 0.80 test accuracy and a consistent decrease in loss from 0.78 to 0.53, until 25 clients and then a marginal drop in accuracy to 0.84 and 0.78 of test accuracy and a marginal increase in loss to 0.56 as represented in Figure 9. FedSGD shows a low and quite inconsistent performance concerning the other two approaches of federated learning. It shows a significant inconsistency in the accuracy and loss change with accuracy, as in Figure 6, ranging from 0.29 to 0.41, and the loss demonstrated in Figure 7 between 1.59 and 1.91.

TABLE 3. Comparison of Accuracy and Loss of different approaches.

No. of Clients	FedAvg		FedSGD		Unweighted FedAvg		CNN	
	Accuracy	Loss	Accuracy	Loss	Accuracy	Loss	Accuracy	Loss
5	0.73	0.81	0.32	1.59	0.74	0.78	0.77	0.94
10	0.74	0.85	0.29	1.89	0.79	0.74		
15	0.76	0.83	0.34	1.91	0.79	0.72		
20	0.79	0.74	0.41	1.84	0.81	0.66		
25	0.82	0.63	0.38	1.84	0.85	0.53		
30	0.82	0.64	0.39	1.82	0.84	0.56		

TABLE 4. Comparison of the Test Accuracies of different approaches.

No. of Clients	FedAvg	FedSGD	Unweighted FedAvg	CNN
5	0.68	0.27	0.69	0.69
10	0.69	0.24	0.75	
15	0.71	0.30	0.71	
20	0.74	0.36	0.74	
25	0.78	0.33	0.80	
30	0.77	0.35	0.78	

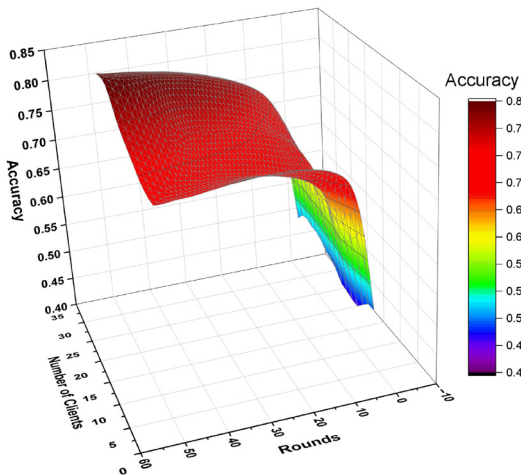


FIGURE 4. The accuracy analysis of the FedAvg approach while varying the number of rounds and the number of clients used for the analysis.

The Figure 3 demonstrates a quite inconsistent graph of accuracy and loss with best accuracy of 0.78 and minimum loss of 0.9254. CNN was trained for 10 epochs with the highest accuracy of 0.77 and 0.94 loss, and also reported a testing accuracy of 0.69 as shown in Table 4.

The results establish that while CNN demonstrated significant stability, the federated learning approaches can achieve better results when data is accessed from multiple devices. It establishes Unweighted FedAvg as most effective approach achieving highest Accuracy and lowest Loss. FedAvg shows a quite consistent change of accuracy and loss, which is always outperformed by Unweighted FedAvg. In contrast, FedSGD demonstrates limited utility due to its subpar performance in both metrics.

VI. DISCUSSION

The results above emphasize the importance of adapting federated learning approaches in video surveillance systems.

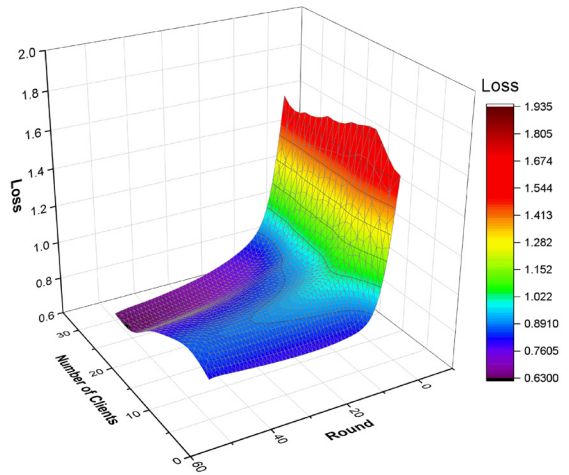


FIGURE 5. The Loss analysis of the FedAvg approach while varying the number of rounds and the number of clients used for the analysis.

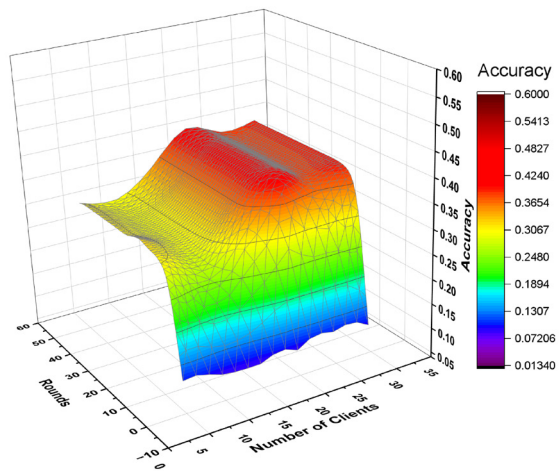


FIGURE 6. The accuracy analysis of the FedSGD approach while varying the number of rounds and the number of clients used for the analysis.

To address R4 Unweighted FedAvg outperformed the two approaches implemented, i.e., FedAvg and FedSGD. It suggests that uniform contribution may enhance the performance of the system. FedAvg also performed significantly, but FedSGD showed inconsistency in accuracy and loss with each round, making it highly unsuitable for implementation in real-world scenarios. Experiments were conducted on a varying number of clients i.e. 5, 10, 15, 20, 25, and 30 To

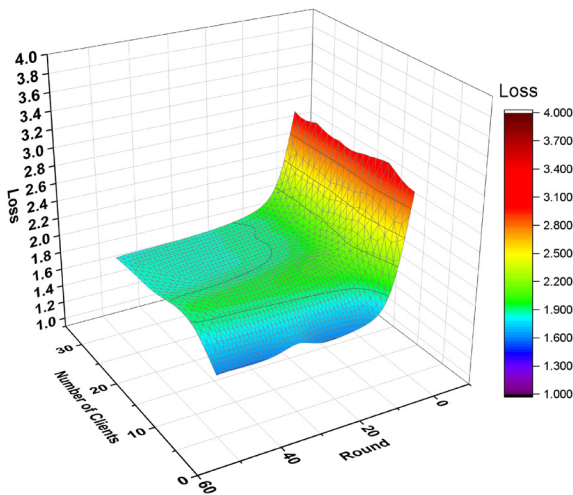


FIGURE 7. The Loss analysis of the FedSGD approach while varying the number of rounds and the number of clients used for the analysis.

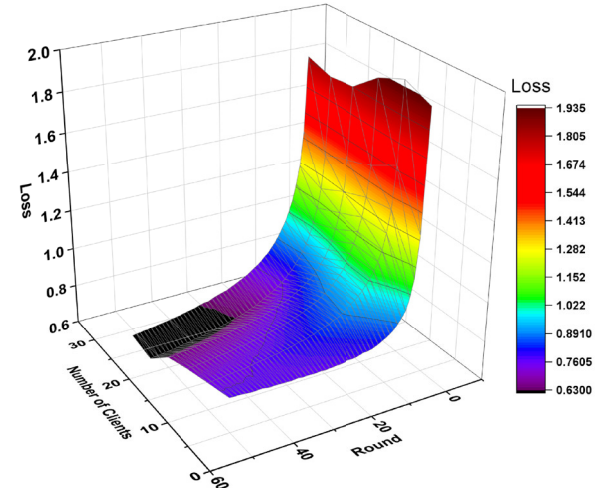


FIGURE 9. The Loss analysis of the unweighted FedAvg approach while varying the number of rounds and the number of clients used for the analysis.

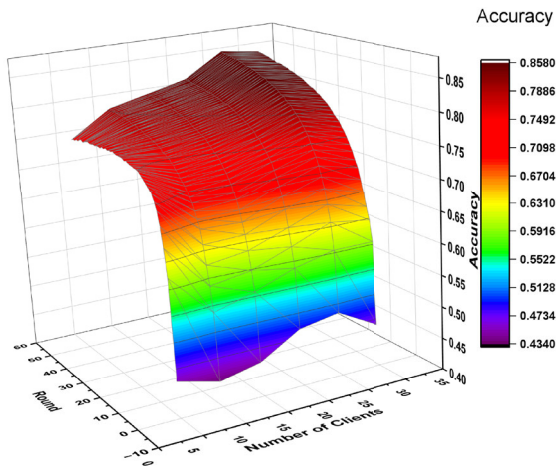


FIGURE 8. The accuracy analysis of the unweighted FedAvg approach while varying the number of rounds and the number of clients used for the analysis.

address R3. With an increase in the number of clients, both FedAvg and Unweighted FedAvg demonstrated consistent improvement in performance, enhancing the model's generalization. Although Unweighted FedAvg experienced a slight drop in accuracy and an increase in loss for 30 clients, it still performs better than FedAvg and FedSGD. Regarding R2, the analysis of training loss trends reveals that both FedAvg and Unweighted FedAvg exhibit steady convergence. The models effectively minimized training loss, indicating efficient local training and successful global aggregation. FedSGD's inconsistent behavior, however, resulted in higher and erratic loss patterns. As far as CNN is concerned, though it performed surprisingly well but it also showed some inconsistency after 7 epochs with continuous fluctuations in accuracy and loss. The federated learning approaches performed better while reducing the communication overhead and preserving privacy. Answering R1 and R4 overall, the Unweighted FedAvg performs best as compared to other approaches in

terms of accuracy and loss, but FedAvg was able to maintain consistency. FedSGD's instability limits its practicality. Scalability and communication efficiency make federated learning approaches a potential alternative to centralized systems.

As our results show promising performance gains using FL over centralized learning, real-world deployments often involve millions of edge devices with limited compute, memory, and battery capacity. Scaling FL to such large environments while ensuring convergence and minimal latency is non-trivial. One of the key bottlenecks is the size of the model parameters and the frequency of communication rounds. Even with FL's reduction in raw data transmission, repeated exchanges of large model weights can impose significant overhead on edge devices. To manage this, techniques such as model compression, sparsification, and quantization of model updates can be employed to reduce communication load. Additionally, scalability can be improved through hierarchical FL architectures, where intermediate aggregators collect updates locally before forwarding them to a global server.

The experiments assumed full client participation across all rounds. The results, particularly the instability observed in FedSGD, indirectly reflect the effects of client heterogeneity and potential drift. FedAvg and Unweighted FedAvg showed greater robustness under non-IID conditions. Fairness and update balance were partially addressed through Unweighted FedAvg, which avoided overweighting clients with more data. Future work can further investigate dynamic client participation, dropout resilience, and personalized FL strategies to enhance fairness and model stability.

While our results show promising performance gains using FL over centralized learning, real-world deployments often involve millions of edge devices with limited compute, memory, and battery capacity. Scaling FL to such large environments while ensuring convergence and minimal latency is non-trivial. One of the key bottlenecks is the size of the model

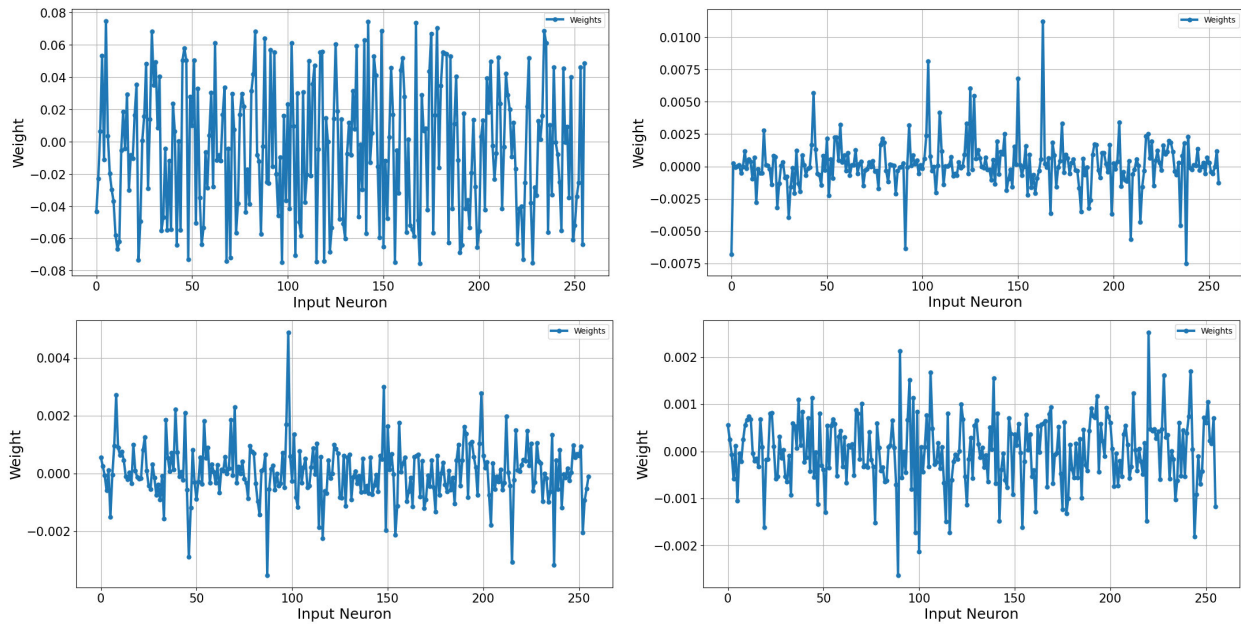


FIGURE 10. A visualization of the updates in the weights.

parameters and the frequency of communication rounds. Even with FL's reduction in raw data transmission, repeated exchanges of large model weights can impose significant overhead on edge devices. To manage this, techniques such as model compression, sparsification, and quantization of model updates can be employed to reduce communication load. Additionally, scalability can be improved through hierarchical FL architectures, where intermediate aggregators collect updates locally before forwarding them to a global server. Federated learning offers inherent privacy benefits by keeping raw data local to edge devices, recent studies have demonstrated that model updates may still leak sensitive information under certain attack scenarios such as model inversion or gradient leakage. This is particularly critical when deploying FL systems in sensitive domains like healthcare or finance, where compliance with privacy regulations (e.g., GDPR, HIPAA) is mandatory. Future work will involve the integration of privacy-enhancing techniques such as differential privacy, homomorphic encryption, or secure multiparty computation to ensure rigorous protection of client data. Additionally, we plan to assess the trade-offs between model utility and privacy in such constrained environments.

Any real-world implementations of the systems should always consider the general practical issues, such as device failures, varying data quality across clients, and dynamic client participation due to heterogeneity issues, including different computational powers, network speeds, battery capacities, and so on. In practical FL systems, clients exhibit significant variability in compute power, network reliability, and energy availability. This system heterogeneity often leads to stragglers, delaying aggregation and reducing training efficiency. Asynchronous Federated Learning (AFL) is one

solution where client updates are integrated into the global model as they arrive, eliminating synchronization bottlenecks. In this context, Federated Primal-Dual Optimization (FedPD) and its variants have emerged as effective algorithms that decouple local updates from strict synchronization requirements. FedPD leverages the primal-dual structure of the optimization problem, allowing clients to update their local models asynchronously while maintaining convergence guarantees even under heterogeneous conditions. Its adaptive versions, such as FedAsyncPD, further improve performance by dynamically weighting updates based on client staleness or reliability, making them highly suitable for deployments with diverse hardware and connectivity profiles. Future research could integrate asynchronous or semi-synchronous learning approaches, such as FedPD and its adaptive variants, to better accommodate the challenges posed by client-side system heterogeneity.

Profiling the system's performance in terms of training time, model size, and memory usage on actual edge hardware, such as smartphones or Raspberry Pi clusters can help to evaluate the practicality of deploying the proposed federated learning approaches in resource-constrained environments. Additionally, we intend to explore lightweight federated learning frameworks such as TensorFlow Federated, and investigate optimization techniques, including FedLite and adaptive sparsification protocols to enhance communication efficiency and scalability.

VII. CONCLUSION AND FUTURE WORK

The study elaborates on the effectiveness of the federated learning approaches for Anomaly detection in video surveillance. Out of the three approaches implemented,

Unweighted FedAvg proves to be the most efficient approach, outperforming FedAvg and FedSGD. Future research could incorporate statistical variance indicators such as standard deviation or confidence intervals to better evaluate model stability across multiple runs and under stochastic or real-world deployment settings.

Federated Learning (FL) involves frequent communication between clients and the central server, where model updates are transmitted after every local training round. This creates a communication bottleneck, increasing energy consumption and slowing down global model updates, especially as the number of clients or model size grows. Although strategies like quantization or sparsification have been proposed to reduce the size of transmitted updates, they do not address the issue of sending potentially redundant or low-impact information. Event-Driven Federated Learning (EDFL) offers a more efficient solution by allowing clients to communicate only when significant local changes occur, such as performance drops or data distribution shifts. This selective communication reduces bandwidth usage and energy consumption while maintaining model performance, making EDFL a scalable and robust alternative in dynamic or resource-constrained environments.

While the study specifically targets suspicious activity detection in public spaces using the UCF-Crime dataset, we recognize that anomaly detection is a critical task across multiple domains such as smart cities, healthcare, and industrial monitoring and our federated learning approaches were evaluated in the context of public surveillance, the core methodology is domain-agnostic and can potentially be adapted to other anomaly detection tasks, provided appropriate domain-specific datasets are available. Future work could involve extending the current framework to these areas by leveraging datasets such as AI City Challenge (for smart traffic surveillance), MIMIC-CXR (for medical anomalies), or MVTec AD (for industrial defect detection). Such extensions would not only help assess the robustness and generalizability of the proposed methods but also expand their applicability in privacy-sensitive environments beyond public security systems.

Future work will explore optimizing aggregation techniques to reduce update variation and global model drift, along with adaptive client selection based on performance criteria. Additionally, a detailed ablation study is planned to isolate the effects of factors like model architecture, update frequency, and client selection on convergence and accuracy. While this paper focuses on aggregation under non-IID settings, deeper analysis of these components can provide valuable insights for improving federated learning efficiency.

In parallel, integrating continual learning principles into the federated setup represents another promising avenue. Surveillance environments often evolve over time, necessitating models that can adapt to new data distributions without forgetting past knowledge. Incorporating federated continual learning frameworks—such as those based on knowledge distillation or collaborative replay—can enable

continuous adaptation while preserving data locality and user privacy. This would be particularly beneficial in real-time or long-term deployments, where the system must remain robust under concept drift and seasonal variation. Another important avenue for future research involves addressing fairness and bias in federated learning systems. In practical deployments, disparities in client participation frequency, data distribution, or resource availability can lead to models that perform disproportionately better for certain users while disadvantaging others—particularly minority or underrepresented groups. Our current use of Unweighted FedAvg attempts to reduce such imbalance by treating all client updates equally, but further fairness-aware techniques—such as participation tracking, re-weighting, or personalized model heads—will be necessary to ensure equitable performance across heterogeneous user populations.

Federated Learning environments show great potential in optimizing the task of Anomaly Detection in Video Surveillance systems. By refining the aggregation techniques, utilizing complementary hardware resources, and addressing the major limitations of its adaptation in real-time large-scale environments, federated learning can successfully improve the current anomaly detection scenarios.

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real-world challenges in multiple areas, combining theoretical rigor with practical impact.

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